UE22CS341A: Software Engineering
Case Study

Unit 1 Deliverable

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Software Requirements Specification (SRS) for Student Mental Health Support System

1. Introduction

1.1 Purpose

This document specifies the requirements for the Student Mental Health Support System. The system aims to provide students with secure access to mental health resources, enable the scheduling and management of counseling sessions, and offer personalized wellness tips. It will allow administrators to monitor overall student well-being through anonymized data analytics, ensuring data privacy and confidentiality.

1.2 Scope

The Student Mental Health Support System is intended for use within a college or university setting, providing a secure and user-friendly platform for managing student mental health services. The system will support students, counselors, and administrators by offering features such as appointment booking, access to mental health resources, wellness surveys, and personalized notifications. It will integrate with the institution's existing infrastructure, including authentication systems and databases, while complying with relevant privacy regulations.

1.3 Definitions, Acronyms, and Abbreviations

- **SRS:** Software Requirements Specification
- **DBMS:** Database Management System
- **UI:** User Interface
- **API:** Application Programming Interface
- **SSO:** Single Sign-On
- **SSL/TLS:** Secure Sockets Layer/Transport Layer Security

- **OAuth:** Open Authorization (protocol for secure authorization)
- **JWT:** JSON Web Token (used for secure user authentication)
- **GDPR:** General Data Protection Regulation
- **HIPAA:** Health Insurance Portability and Accountability Act (relevant for data protection)

1.4 References

- "Development of a Smart Mental Health Support System," ScienceOpen. Available: ScienceOpen Document
- "How to Design a Database for Healthcare Management System," GeeksforGeeks. Available: GeeksforGeeks Article

1.5 Overview

This document is structured into sections detailing the functional and non-functional requirements, system features, external interface requirements, and other necessary details for the development of the Student Mental Health Support System.

2. Overall Description

2.1 Product Perspective

The Student Mental Health Support System is part of the institution's existing IT infrastructure, interfacing with the university's database systems and Single Sign-On (SSO) services. The system will provide secure, reliable access to mental health resources and support services for students, counselors, and administrators.

2.2 Product Functions

- **User Authentication and Authorization:** Secure login with role-based access control.
- **Counseling Session Management:** Booking, calendar integration, session notes, and follow-up reminders.
- **Resource Database:** Repository of mental health articles, videos, and guides.
- **Wellness Survey and Personalized Tips:** Periodic surveys and algorithm-driven wellness tips.
- **Anonymized Data Analytics Dashboard:** Reports on mental health trends and resource usage.

2.3 User Classes and Characteristics

- **Students:** Access mental health resources, book counseling sessions, complete wellness surveys, and receive personalized tips.
- **Counselors:** Manage appointments, access student records, and update treatment plans.
- **Administrators:** Monitor system usage, manage resources, and generate anonymized reports.

2.4 Operating Environment

- **Software:** Web application compatible with modern browsers, running on secure servers.
- **Hardware:** Servers for hosting the application and database, client devices (PCs, tablets, smartphones) for user access.
- **Network:** Secure, encrypted connections to the university's network and external services.

2.5 Design and Implementation Constraints

- **Compliance:** The system must comply with GDPR, HIPAA, and other relevant data protection laws.
- **Security:** Must use encryption for data at rest and in transit, and provide secure authentication mechanisms.
- **Usability:** The system should be intuitive, user-friendly, and accessible to all users.

2.6 Assumptions and Dependencies

- The system assumes the availability of a stable network connection and regular maintenance.
- Integration with the university's SSO system is available.
- Users have access to devices that can run modern web browsers.

3. External Interface Requirements

3.1 User Interfaces

- **Web-based UI:** Responsive design for desktops, tablets, and smartphones.
- **Dashboard:** Customizable interface for administrators to monitor usage and trends.
- **Notifications:** System alerts and reminders via email or in-app notifications.

3.2 Hardware Interfaces

- **Servers:** Hosting the application, database, and APIs.
- **Client Devices:** Students, counselors, and administrators will use personal or institutional devices to access the system.

3.3 Software Interfaces

- **APIs:** Integration with the university's SSO system and external services for notifications and data analysis.
- **Database:** Backend database system (MongoDB) for storing user information, appointments, and resources.

3.4 Communication Interfaces

- **Protocols:** Secure communication using SSL/TLS for all data exchanges.

- **Authentication:** OAuth 2.0 or JWT for secure user authentication.

4. System Features

4.1 User Authentication and Authorization

4.1.1 Description

The system will require users to authenticate using secure login credentials, with role-based access controls.

4.1.2 Functional Requirements

- The system shall validate user credentials against the university's SSO system.
- The system shall enforce role-based access, restricting features based on user roles (student, counselor, administrator).

4.2 Counseling Session Management

4.2.1 Description

The system will allow students to book, modify, and cancel counseling sessions, with reminders and notes for counselors.

4.2.2 Functional Requirements

- The system shall allow students to book available time slots with counselors.
- The system shall send notifications and reminders for upcoming sessions.
- The system shall allow counselors to add session notes and schedule follow-ups.

4.3 Resource Database

4.3.1 Description

A comprehensive repository of mental health resources will be available to students.

4.3.2 Functional Requirements

- The system shall provide a searchable database of articles, videos, and guides.
- The system shall allow students to filter resources by topic and type.

4.4 Wellness Survey and Personalized Tips

4.4.1 Description

The system will periodically survey students and offer personalized wellness tips based on their responses.

4.4.2 Functional Requirements

- The system shall periodically prompt students to complete wellness surveys.
- The system shall generate personalized wellness tips based on survey data.

4.5 Anonymized Data Analytics Dashboard

4.5.1 Description

Administrators will have access to a dashboard that provides insights into student mental health trends on campus.

4.5.2 Functional Requirements

- The system shall generate anonymized reports on mental health trends.
- The system shall provide visual analytics, such as heatmaps and charts, for high-stress periods and resource usage.

4.6 Error Handling

4.6.1 Description

Handles errors such as network failures or invalid inputs.

4.6.2 Functional Requirements

- The system shall notify the user of any errors during an operation.
- The system shall log all errors for analysis and troubleshooting.

5. Non-Functional Requirements

5.1 Performance Requirements

- The system shall respond to user inputs within 2 seconds.

5.2 Security Requirements

- The system shall encrypt all user data during transmission using SSL/TLS.
- The system shall enforce multi-factor authentication for administrators.

5.3 Usability Requirements

- The system shall provide an intuitive and user-friendly interface.

5.4 Reliability Requirements

- The system shall have an uptime of 99.9%.
- The system shall regularly back up data to prevent loss.

6. Other Requirements

6.1 Regulatory Requirements

- The system shall comply with local and international regulations, such as GDPR and HIPAA, ensuring data protection and privacy.

6.2 Environmental Requirements

- The system shall operate effectively under a temperature range of 10°C to 40°C, considering both hardware and software components.

7. Requirements Traceability Matrix (RTM)

The RTM will be used to ensure that all requirements are covered by design, development, and testing activities. For example:

Requirement ID	Description	Design Spec	Implementation	Test Case
FR1	User Authentication and Authorization	Section 4.1	Module 1	TC1
FR2	Counseling Session Management	Section 4.2	Module 2	TC2

This SRS document provides a detailed guide for developing the Student Mental Health Support System, ensuring that all functional and non-functional requirements are clearly defined and traceable throughout the project lifecycle.